CSE331: Automata and Computability
Assignment 02

Group Formation: You may form a group of **at most 3 people.**

Deadlines and Submission Links:

— CFG (Question 3):
- Deadline: Dec 12, 2025 11:59 PM
- Submission Link: [To Be Uploaded]
- Submit the Hard copy also.

— Pumping Lemma and Ambiguity (Questions 2 and 5):
- Deadline: Dec 19, 2025 11:59 PM
- Submission Link: [To Be Uploaded]
- Submit the Hard copy also.

— Question 01, Pushdown Automata, Turing Machine (Questions 4, 6):
- Deadline: Dec 27, 2025 11:59 PM
- Submission Link: [To Be Uploaded]
- Submit the Hard copy also.

We will follow the deadline strictly. After the deadline ends, I will share the solution, so no submissions will be accepted after the solution is provided.

[This doc file contains both the Assignment and Practice Sheet [Click Here] on the final syllabus. The solution for the practice sheet can be found [Click Here]] *[Self Note: Link should be updated - Done]*

Solve all the questions given below:

1) List five use cases where and how you can apply the knowledge from our CSE331 course.

2) Proof L is a non-regular language using the Pumping Lemma:
   a) $L = \{w \in \{0,1\}^*: \ w = 0^{n!}, \ n \geq 0\}$
   b) $L = \{w \in \{0,1\}^*: \ w = 0^n 1^n 1, \ n \geq 0\}$

c) $L = \{w \in \{0,1\}^*: w = 0^a1^b1^c0^d$, where $a + b = c + d$ and $a, b, c, d \geq 0\}$ [Be careful for (b): check if xyyz works]

d) $L = \{w \in \Sigma^*: w = a^ib^j$, where $i > j$, and $j \geq 0\}$

e)

3) Design a Context Free Grammar for the Language:

a) $L = \{w \in \{a,b,c,p,q,r,\#\}^*: a^i\#^nc^jp^{2x}q^yr^zb^k$ where $i=j+k$, $y=3x+z$, z is even, n is odd, and $i,j,k,n,x,y,z \geq 0\}$

b) $L = \{w \in \{0,1,2\}^*: w = 0^i2^j1^k$, [where ……conditions……] }

where…

i) $i = k$, $i,k \geq 1$ and $j \geq 2$

ii) $i = 3k$, j is odd, and $i,j,k \geq 0$

iii) i is a multiple of two, k is two more than a multiple of 3, $j = k+i$, and $i,j,k \geq 0$

iv) $i+j > k$ and $i,j,k \geq 0$

v) $i+k$ is even, $j = i+k$ and $j \geq 1$

c) $L = \{w \in \{0,1\}^*$: the parity of 0s and 1s is different in w}

d) $L = \{w \in \{0,1\}^*$: the number of 0s and 1s are different in w}

[Hint: First, try to solve for an equal number of 0s and 1s in w]

e) $L = \{w \in \{a, b\}^*$: w is a palindrome}. Design a CFG for $\overline{L}$

f) $L = \{w \in \{a, b\}^*$: w does not contain any palindromic substring of length two}

g) $L = \{w\#x \mid w^R$ is a substring of x where $w \in \{0, 1\}^*\}$

h) $L = \{1^i02^j1^k \mid i, j, k \geq 0, 3i \geq 4k + 2$, j is not divisible by three}

i) Recall that for a string w, $|w|$ denotes the length of w. $\Sigma = \{0,1\}$

$L1 = \{w \in \Sigma^*$: w contains exactly two 1s}

$L2 = \{x\#y : x \in \Sigma^*, y \in L1, |x| = |y|\}$

Construct a CFG for L2.

j) Recall that for a string w, |w| denotes the length of w. $\Sigma = \{0,1\}$

$L1 = \{w \in \Sigma^*: \text{w contains at least three 1s}\}$

$L2 = \{x\#y : x \in (\Sigma\Sigma)^*, y \in L1, |x| = |y|\}$

Construct a CFG for L2.

k) Recall that for a string w, |w| denotes the length of w. $\Sigma = \{a,b\}$

$L1 = \{w \in \Sigma^*: \text{w contains bb as a substring}\}$

$L2 = \{w1\#w2 : w1, w2 \in L1 \text{ and } |w1| = |w2|\}$

Construct a CFG for L2.

4) For all the problems in Question(3), now construct the Pushdown Automata (PDA).

5) **Derivation, Parse Tree, Ambiguity**

i) Given the Context-Free Grammar, answer the following questions

$$A \rightarrow A1 \mid 0A1 \mid 01$$

(a) Give a leftmost derivation for the string 001111.
(b) Sketch the parse tree corresponding to the derivation you gave in (a).
(c) Demonstrate that there are two more parse trees (apart from the one you already found in (b)) for the same string.
(d) Find a string w of length six such that w has exactly one parse tree in the grammar above. (1 point)

ii) Given the Context-Free Grammar, answer the following questions:

$$A \rightarrow 1A \mid 1C \mid 0B \mid 00A$$
$$B \rightarrow 0A \mid 1B \mid 00B$$
$$C \rightarrow 0C0 \mid 0C1 \mid 1C0 \mid 1C1 \mid \varepsilon$$

(a) Give a leftmost derivation for the string 01011001. (3 points)
(b) Sketch the parse tree corresponding to the derivation you gave in (a). (2 points)

(c) Demonstrate that the given grammar is ambiguous by showing two more parse trees (apart from the one you already found in (b)) for the same string. (3 points)
(d) Find a string w of length six such that w has exactly one parse tree in the grammar above. (1 point)
(e) Design an unambiguous Context Free Grammar for the language represented by the given ambiguous grammar. (1 point)

6) Turing Machine

a) L = $\{1^{3^n} \mid n \geq 0\}$
b) L = {w ∈ {0,1,2}*:$0^i1^j2^k$ where i=j=k and i,j,k ≥ 0}
c) L = {w ∈ {0,1,2}*:$0^i\#1^j\#2^k$ where i*j=k and i,j,k ≥ 0} [Idea described in Sipser's Book]
d) L = {w ∈ {a,b}*: w is an odd length palindrome}
e) L = {$w\#w^R\#w$ | w ∈ {a,b}*} [w^R means reverse of w]

CSE331: Automata and Computability
Practice Sheet
Prepared By: KKP

The solution for the practice sheet can be found [Click Here]
[Self Note: Link should be updated - Done]

[There are also some practice problems in the handnotes]

Question 1: Designing Context-Free Grammar

Give Context-free Grammar that generates the language

a) L = {w ∈ {0,1}*: w starts with '**0**' and the length of w is even.}
b) L = {w ∈ {a,b}*: every second letter in w is a '**b**'.}
c) L = {w ∈ {a,b}*: the length of w is divisible by three}
d) L = {w ∈ {0,1}*: w starts with and ends with a different symbol.}
e) L = {w ∈ {a,b}*: the number of '**a**' is at least the number of '**b**' in w.}
[**Hint**: Link]
f) L = {w ∈ {a,b}*: the length of w is two more than a multiple of six}
g) L = {w ∈ {0,1}*: w = 0^n1^{2n+3}, n ≥ 0 }
h) L = {w ∈ {0,1}*: w = 0^n1^n, where n is odd.}

i) L = {w ∈ {0,1,2}*: $w=0^i1^j2^k$ where j ≥ 2i + 3k, and i, k ≥ 0}

j) L = {w ∈ {0,1,2,3}*: $w=0^i1^j2^k3^m$ where i=m, j ≥ 3k+2, and m, k ≥ 0}

k) L = {w ∈ {0,1,2}*: $w=0^i1^j2^k$ where i > 2j + 3k, and j, k ≥ 0}

l) If A = {w ∈ {0, 1}* : w contains at least two 0s}, then construct L = {w ∈ {0, 1}*: w = $0^{3i}v1^{2i}$ where v ∈ A and i ≥ 0} [marked as problem j in the solution]

m) Recall that for a string w, |w| denotes the length of w. Σ = {0,1}
   L1 = {w ∈ Σ^*: w contains 11}
   L2 = {x#y : y ∈ L1, x ∈ Σ^*, |x| = |y|}
   Construct a CFG for L2.

n) L = {w ∈ {0,1}* : w_1#w_2: number of 0s w_1 is equal to number of 1s in w_2.}

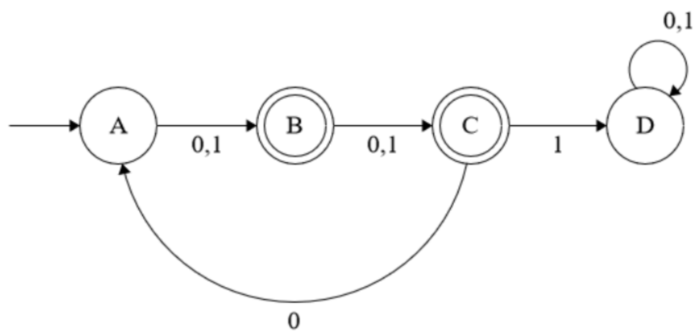
o) L = {w ∈ {0,1}* : w_1#w_2: length of w_2 is double of length of w_1.}

Construct a Context-free Grammar for the language expressed by the following Finite Automata:

a) Convert the Regular Expression into a CFG: ((ab)* + (a+b^3) cb)*

b) Convert the Regular Expression into a CFG: (a*b* + (ac+b*c) b)*

c) Convert the following DFA into a CFG:



Question 2: Derivation, Parse tree, Ambiguity

I.

$$S \rightarrow ASB \mid SS \mid SAS \mid A$$
$$A \rightarrow ASS \mid BS \mid B$$

B→00|11|01|1

Given the above Context-Free Grammar, answer the following questions:

a) Show the Leftmost Derivation of 00010111
b) Sketch the parse tree corresponding to the derivation you gave in (a).
c) Show the Rightmost Derivation of 00010111
d) Sketch the parse tree corresponding to the derivation you gave in (c).

II. S → 0S1 | 1S0 | SS | 01 | 10

Given the above Context-Free Grammar, answer the following questions:

a) Show that the grammar above is ambiguous by demonstrating two different parse trees for 011010.
b) Find a string w of length six such that w has exactly one parse tree in the grammar above.

Question 3: Pushdown Automata

a) L = { w ∈ {0,1}* : w = 0^n1^n, where n is odd.}
b) L = { w ∈ {0,1}* : w = 0^m1^n, where m,n ≥ 1 and m≥n.}
c) L = { w ∈ {0,1}*: w | the length of w is divisible by four}
d) L1 = { w ∈ {0,1}*: the number of 1s in w is multiple of 3.}
L2 = { w ∈ {0,1}* : w contains even numbers of 0.}
Construct a PDA for L = { w ∈ {0,1}*: w = uv, where u ∈ L1,v ∈ L2 and |u| = |v| }
e) L = { w ∈ {0,1}* : w = $0^i1^j0^k$, where j = i+k and i, k ≥ 0.}
f) L = { w ∈ {0,1}* : w = $0^i1^j0^k$, where i+j = k and i, k ≥ 0.}
g) L = { w ∈ {0,1}* : w_1#w_2: number of 00 substrings in w_1 is equal to number of 11 in w_2.}
h) L = { $w\overline{w}^R$: w ∈ {0,1}* }

$$L = \{w\overline{w}^{\mathcal{R}} : w \in \{0,1\}^*\}$$

Here, $\overline{w}^R$ denotes the reverse complement of the string w.

For example, $01001101 \in L$, because the second half of the string, 1101, is the reverse complement of the first half, 0100, i.e. $1101 = \overline{0100}^R$

Question 4: Chomsky Normal Form - CNF [Not included in the Final Syllabus]

a) Convert the following grammar into Chomsky Normal Form. You must show work.

$$S \rightarrow bBS \mid \varepsilon$$
$$B \rightarrow bYb \mid bY$$
$$C \rightarrow Ccb \mid c \mid A$$
$$Y \rightarrow bcY \mid \varepsilon \mid YBF$$

Here b, and c are terminals and the rest are variables.

b) Convert the following grammar into Chomsky Normal Form. You must show work.

$$S \rightarrow YaSb$$
$$X \rightarrow aXY \mid bX \mid Y$$
$$Y \rightarrow X \mid b \mid \epsilon$$

Here a, and b are terminals and the rest are variables.

c) Convert the following grammar into Chomsky Normal Form. You must show work.

$$S \rightarrow bXaY \mid ZXb$$
$$X \rightarrow aY \mid bY \mid Y$$
$$Y \rightarrow X \mid c \mid \varepsilon$$
$$Z \rightarrow ZaX$$

Here a, b, c are terminals and the rest are variables.

d)

Problem 3 (CO2): Chomsky Normal Form (10 points)

(a) **List** the rules that violate the conditions of Chomsky Normal form in the following grammar. Here a, b, and c are terminals and the rest are variables.

$$A \to BC \mid \mathtt{b}B \mid \mathtt{a}$$
$$B \to \mathtt{bb} \mid C\mathtt{b} \mid \mathtt{b} \mid C$$
$$C \to \mathtt{c}$$

(b) **Write** down the additional rules that need to be added to the following grammar if the production, $B \to \varepsilon$ is removed. Here 0 and 1 are terminals and the rest are variables.

$$S \to AB \mid 1$$
$$A \to BAB \mid ABA \mid B \mid 11$$
$$B \to 00 \mid \varepsilon$$

(c) **Write** down the additional rules that need to be added to the following grammar if the unit productions are removed. Here 0 and 1 are terminals and the rest are variables.

$$S \to XYX \mid YX \mid X \mid Y$$
$$Y \to XY \mid X0 \mid 0$$
$$X \to 1 \mid Y$$

Question 5: CYK Algorithm [Not included in the Final Syllabus]

a) Consider the following grammar in CNF form.

S→BC | CD
B→CB | c
C→DD | c
D→BC | b

Show if the string w=cccbb can be derived from the grammar above.

b) Apply the CYK algorithm to determine whether the string "abcaa" can be derived in the following grammar. You must show the entire CYK table. Here a, b, c are terminals and the rest are variables.

S → CA
A → AA | AD | a
B → AB | BC | b
C → CA | BC | c
D → a